

# Juan David Munoz-Bolanos

## Biomedical Optical Engineer | Medical Imaging & Vision



### Science

Rechenauerstraße 42, Innsbruck, Austria

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## PROFESSIONAL SUMMARY

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Biomedical engineer based in **Innsbruck** with hands-on **Medical optical microscopy** and **Computer Vision**. Hands-on confocal, spectroscopy and two-photon imaging for clinical applications, and analytical image processing pipelines.

## TECHNICAL COMPETENCIES

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### Biomedical Optics & Medical simulation

Optical desing - End-to-end optical system modeling by numerical estimations and Zemax OpticStudio, including tolerance analysis. Optical prototyping experience, following for example ISO 10110. Adaptive optics for optical aberration correction.

### Vision Technology & Digital Metrology

Image Processing Pipelines using tailored **Python** pipelines, and computational/holography imaging with GPU Acceleration (Jax/Cupy).

### Hardware Control & Medical Systems

Biomedical imaging for deep tissue imaging using infrared light. Embedded control knowledge using FPGA and optomechanical devices.

### Languages

English (Proficient), German (Intermediate/Working Proficiency), Spanish (Native).

## PROFESSIONAL EXPERIENCE

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### Medical University of Innsbruck

*PhD Researcher - Optical Systems Engineering*

Innsbruck, Austria

*Jan 2022 – Present*

- **Biomedical Optics & Adaptive Optics:** Designed control system for the optical microscopes to correct optical aberrations and thus imaging deeper in biological tissue.
- **Algorithmics & GPU Acceleration:** Image processing of phase aberrations and GPU numerical estimations of image enhancement.
- **Optics Lab Management:** Responsible for the maintenance and improvement of the microscopy lab, including handling fine optical components, ensuring operational standards in biomedical research.

### Leibniz Institute of Photonic Technology (IPHT)

*R&D Assistant Researcher*

Jena, Germany

*Jun 2020 – Dec 2021*

- **Prototyping:** Medical Imaging Prototyping: assembling multimodal imaging strategies for bladder cancer identification.
- **System Integration:** Collaboration with collegeus to create data analysis strategies, such as machine learning to detect the cancer.

### INTECOL S.A.S.

*Computer Vision Engineer*

Medellin, Colombia

*Mar 2018 – Mar 2019*

- **Digital Measurement & Reliability:** Development cycle of quality control systems using computer vision by means of industrial camera, illumination and processing data algorithms.
- **Safety-Critical Inspection:** Manage the product lifecycle from concept to industrial validation, ensuring reliability in safety-critical public areas.

## EDUCATION

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### Medical University of Innsbruck

*Ph.D. Biomedical Microscopy & Optics (Exp. March 2026)*

Austria

*Jan 2022 – Present*

### Friedrich Schiller University Jena

*M.Sc. in Photonics*

Germany

*2019 – 2021*

### University of Cauca

*B.Sc. in Physics Engineering*

Colombia

*2012 – 2018*

## SELECTED PUBLICATIONS & AWARDS

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- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. *Applied Sciences*, 14(11), 4726.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 2025, 12 (1), 176–184.

## REFERENCES

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### Prof. Monika Ritsch-Marte

*Head of Institute for Biomedical Physics*

[monika.ritsch-marte@i-med.ac.at](mailto:monika.ritsch-marte@i-med.ac.at)

Medical University of Innsbruck

### Prof. Christoph Krafft

*Group leader: spectroscopy imaging & Histopathology*

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Leibniz Institute of Photonic Technology